Post-root lift and check audit (rev2, 2026-10-27)

Class legend: D = derived from source and weights, R = recomputed from recorded rows, E = estimate, T = theorem proved elsewhere. No experiment was run; HEAD fbccdb89e; hashes of the six sources are in `costs.json`. This is not a cross-method comparison, so the 0/0.1/1 GAE band does not apply. ECC2K-130 (n=131) appears in no row; shares at n>=83 are count-model extrapolation.

1 Question

Which steps after a root or decomposition is found (lift, y-recovery, rationality or trace test, certificate) are removable by theorem T2 (the twist lift is degenerate only) or are redundant by group closure, and what do they cost against the step they sit in?

2 Findings

1. (D, R) P1 incumbent PairTable n=47: scan is 97.35% of per-relation bitop; hit recovery 1.29%, independent re-verification 1.32%, fast re-add 0.046%, finish_lazy 0.003%. The path has no root, y-recovery or rationality step. Nothing for T2 to remove.
2. (D) P3 n=83 S3 path: the trace test (about 166 bit-ops, 0.108% of a probe) is dead on E(F_q) abscissae by group closure. The binding constraint on the exceptional path is the start-up assert at `koblitz_orbit_dlp_fast.rs` about 6321 (solver roots equal library `s3_x_roots`), not the trace test: `finish` returns to `s3_x_roots` first.
3. (D, corrected) P4 `lift_candidate` (7301) calls the generic `points_with_x` (1374), which calls `solve_artin_schreier` (`binary_semaev.rs` 191): a full absolute trace first, then the half-trace. At n=47, bitop/v1: trace pass 46 S + 46 A = 46*184 + 46*47 = 10,626 units. The y-recovery is 56,160 with that pass (45,534 is the `bitop_tally` comment, which omits it). The trace pass is 18.9% of y-recovery (23.3% of the tally figure). Rev1 reported 231 units, 0.51%: that was `points_with_x_fast`, which this path does not call. Withdrawn. The per-relation share is not computed (solver cost per root and roots per relation are unrecorded at n=47), so no "under 1%" claim and no PROTOCOL decision is made. The pass is a candidate saving (use the half-trace and its final square check, as `points_with_x_fast` does), an engineering item, not a theorem result. The test does reject the degenerate twist family, which is the T2 object.
4. (R, withdrawn anomaly) Rev1 flagged "9 failures in 66 targets" as unexplained. The 66 is `scalar_mults`. Recorded targets are `targets_tried` = 42,811 (solver calls 42,395), so the rate is 9/42,811 = 2.1e-4 per target, the same order as the predicted 1e-4. Mechanism (D, read, not run): `binary_subspace_factor_base` loops `idx in 1..(1<<l)`, so x=0 is never in `x_index`, and `lift_abscissae` returns None at `x_index.get(x)?` before any addition. Failed lifts cost 0 additions, consistent with adds 580 vs 583 and 43,608 vs 43,614. The T2-violation hunt is withdrawn.
5. (R) Recorded wall at n=47, T01-R1, 15 repetitions: collect lap median 7.63 ms, push lap median 1.61 ms (1.54 and 7.42 ms were repetition 0 only); push lap is 6.9 to 8.3% of cold setup (median 7.3%), 17.2% of collect plus push. The lap is `FactorBaseLogSolver::push` (13821): `verify_collected_relation`, de-duplication and `LogSystem::push` (BigUint modpow). The split is unrecorded, so it is an upper bound on verification alone and no "13x undercount" is claimed.
6. (R) P3 recorded: 384 rows, 86,474,156 probes, 295.9 ns per probe, relation check 0.0193% of target time (median 0.023%, max 11.7% on a 949-probe row). P2: 6,252 records, 806,360,309 lookups, 56,006 relations, 0 frobfold mismatches; pooled 14,416 summand adds per relation, but the median row has 96 and the maximum is 25,000,644, so the confirm add is about 1% at the median row and 0.007% only as a pooled artifact.

3 Limits

Pricing omissions are conservative for "small": P1 per-summand price omits canonical key and filter probe; P3 candidate price omits linear-map squarings and half-trace apply. The 52 of 108 n=47 draws that did not lift (uniform in F_q^*, not in V) are the twist half; their share of the select lap (16.7% of cold setup) is not quantified. T2 was exercised only at m<=3 on subspace bases. P5 omits n=23 run-b direct (timed out, seq 6, no cost record), run-b symmetric and pair-table rows, and n=7 run-b rows. The 1% line in figure 2 is a visual aid over mixed denominators (bitop per relation, per probe, per y-recovery; wall percent of cold setup or online; GAE percent), not a common-unit gate. Prediction (labelled): dropping the P4 trace pass changes P4 y-recovery by about 19% and total wall by an amount not estimated here.

4 Weights and commands

bitop/v1: $M=n^2+(n-1)^2+4(n-1)$, $S=4(n-1)$, $A=n$, $I=10n^2$, affine add = I+2M+S+9A. n=47: M 4509, S 184, I 22090, add 31715. n=83: M 13941, S 328, I 68890, add 97847. Re-aggregation was done with throwaway Python over the recorded `price.json`, `*_targets_result.jsonl` and `records.jsonl` files named in `costs.json`; figures regenerate with `python plot.py` in this directory.

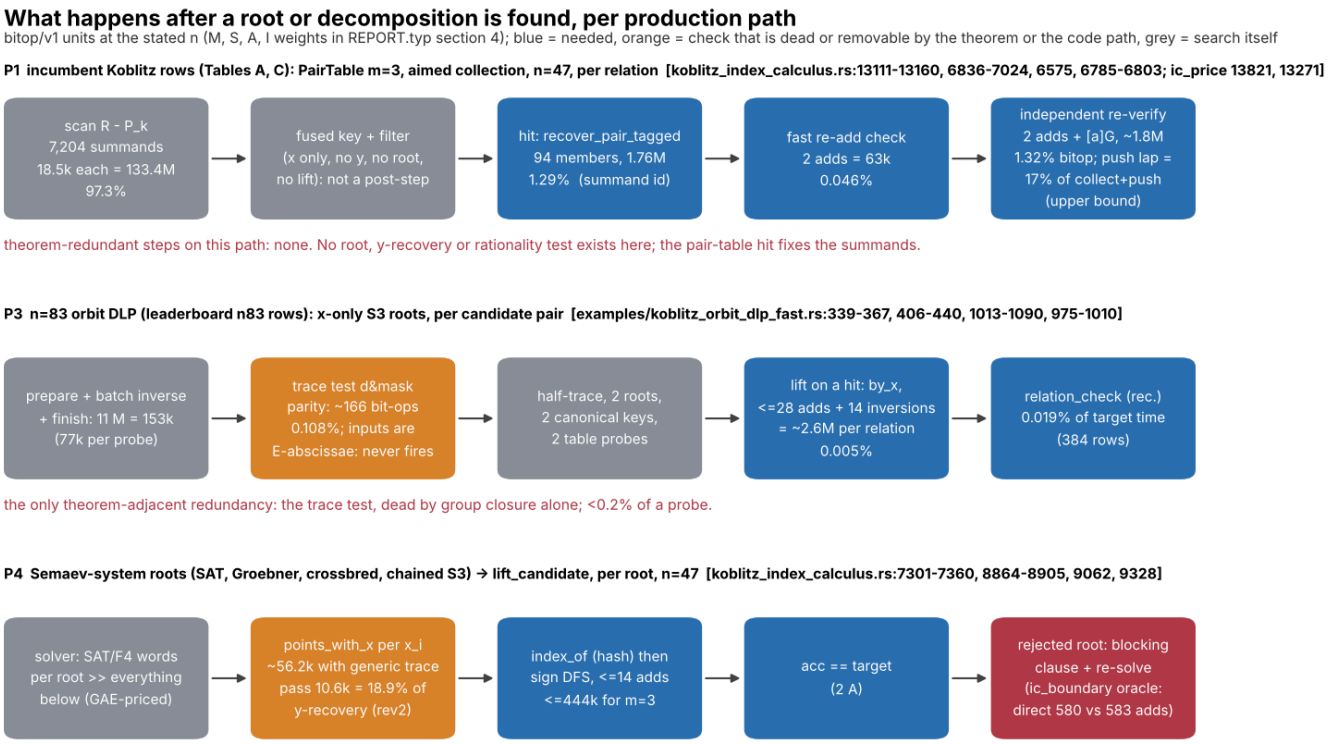


Figure 1: Pipelines per production path.

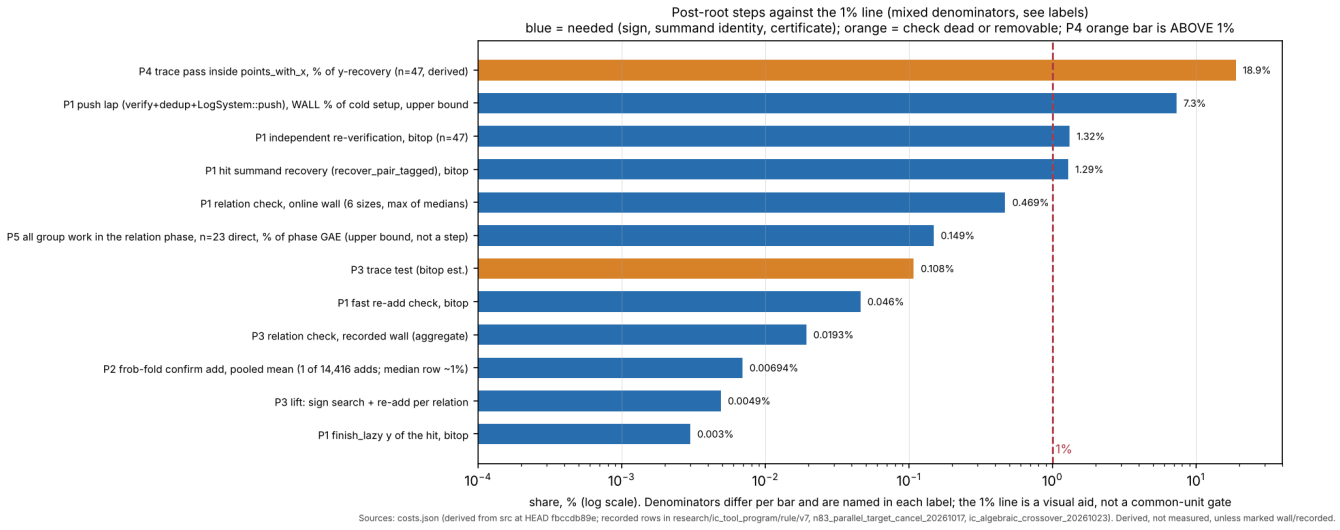


Figure 2: Shares against a 1% visual line; denominators differ per bar.